

Density estimation for mixed Euclidean and non-Euclidean data in the presence of measurement error

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1 The problem

Most deconvolution density estimators are developed for Euclidean data. However, many datasets contain both Euclidean and non-Euclidean variables, such as a real-valued pollutant level and a circular wind direction. This paper develops a deconvolution density estimator for mixed data taking values in

$$\mathbb{R}^d \times \mathbb{G}_c.$$

2 The setup

Let

$$X = (X_1, X_2),$$

where $X_1 \in \mathbb{R}^d$ is Euclidean and $X_2 \in \mathbb{G}_c$ is non-Euclidean. The latent variable X is not observed directly. Instead, we observe

$$Z = (Z_1, Z_2) = (U_1 + X_1, U_2 \circ X_2),$$

where $U_1 \in \mathbb{R}^d$ and $U_2 \in \mathbb{G}_c$ are unobservable measurement errors.

The goal is to estimate the joint density f_X with respect to the product measure

$$\text{Leb}^d \otimes \mu.$$

The assumptions include

$$X \perp U, \quad U_1 \perp U_2,$$

and the measurement error densities f_{U_1} and f_{U_2} are known.

3 The main idea

The estimator combines Fourier deconvolution on $\mathbb{R}^d$ with group Fourier deconvolution on $\mathbb{G}_c$.

For the Euclidean component,

$$Z_1 = X_1 + U_1$$

implies

$$\phi_{Z_1}(t) = \phi_{X_1}(t)\phi_{U_1}(t).$$

Therefore, the Fourier transform of X_1 can be recovered by dividing by $\phi_{U_1}(t)$. This leads to the Euclidean deconvolution kernel

$$K_h^D(s) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} e^{-i\langle t, s \rangle} \frac{\phi_K(t)}{\phi_{U_1}(t/h)} dt.$$

For the non-Euclidean component, Fourier analysis is carried out on the compact Lie group $\mathbb{G}_c$. The model

$$Z_2 = U_2 \circ X_2$$

gives the convolution relation

$$\phi_{Z_2}(\sigma^M) = \phi_{X_2}(\sigma^M) \phi_{U_2}(\sigma^M)$$

for irreducible representations $\sigma \in \widehat{\mathbb{G}}_c$. When $\phi_{U_2}(\sigma^M)$ is invertible, the Fourier coefficients of X_2 can be recovered by multiplication with

$$\phi_{U_2}(\sigma^M)^{-1}.$$

To control variability, the expansion is truncated to representations satisfying

$$k_\sigma < T_n,$$

where k_σ is the Casimir spectrum.

Combining the two corrections gives

$$\widehat{f}_X(x) = \frac{1}{nh^d} \sum_{i=1}^n K_h^D \left(\frac{x_1 - Z_{i,1}}{h} \right) \operatorname{Re} \{ K_{T_n}(x_2, Z_{i,2}) \},$$

where $x = (x_1, x_2) \in \mathbb{R}^d \times \mathbb{G}_c$.

A key property is

$$E \left[K_h^D \left(\frac{x_1 - Z_1}{h} \right) K_{T_n}(x_2, Z_2) \mid X \right] = K \left(\frac{x_1 - X_1}{h} \right) K_{T_n}^*(x_2, X_2).$$

Thus, in conditional expectation, the deconvolution kernels constructed from Z behave as if the latent variable X were observed.

4 The main results

The paper derives pointwise convergence rates, partial uniform convergence rates over $\mathbb{G}_c$, full uniform convergence rates over $\mathbb{R}^d \times \mathbb{T}^q$, L^2 convergence rates over $\mathbb{R}^d \times \mathbb{T}^q$, and asymptotic normality under suitable smoothness conditions. Simulation studies and a real data analysis compare the estimator with naive estimators that ignore measurement error.

5 My takeaways

- Although the estimator uses a product of Euclidean and non-Euclidean kernels, why does it still estimate a general joint density rather than the product of the two marginal densities?
- How should the bandwidth h and the truncation level T_n be chosen together when the Euclidean and non-Euclidean measurement errors have different degrees of smoothness?
- At which step is the assumption $U_1 \perp U_2$ used, and how would the deconvolution relation change if the two measurement errors were dependent?

References

- [1] J. M. Jeon and I. Van Keilegom. Density estimation for mixed Euclidean and non-Euclidean data in the presence of measurement error. *Journal of Multivariate Analysis*, 193:105125, 2023. <https://doi.org/10.1016/j.jmva.2022.105125>.